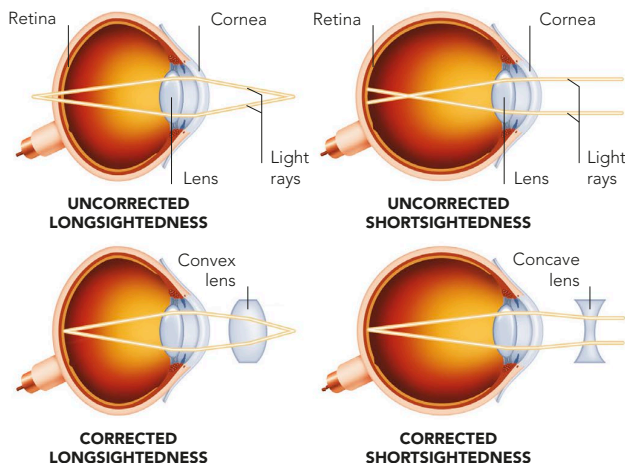


FOCUSING PROBLEMS

Longsightedness (hypermetropia) and shortsightedness (myopia) result from the eyeball being either too short or too long (see right). In astigmatism, vision is blurred because the cornea is irregularly curved, and the lens cannot focus all light rays on the retina. Ageing often affects near vision, as the lens loses its elasticity and cannot easily adjust its shape. Refractive errors can usually be corrected by glasses or contact lenses, or by surgical techniques such as laser-assisted in-situ keratomileusis (LASIK) and photorefractive keratectomy (PRK). In LASIK, the middle layers of the cornea are reshaped by a laser, while in PRK, areas of the cornea's surface are shaved away by a laser to alter its shape.



LONGSIGHTEDNESS

In longsightedness, the eyeball is too short, so the cornea and lens focus light rays behind the retina, and the image is blurred. Convex lenses make the light rays converge so that they are focused on the retina, correcting vision.

SHORTSIGHTEDNESS

In shortsightedness, the eyeball is too long, so the cornea and lens focus light rays in front of the retina, and the image is blurred. Concave lenses are required, which make the light rays diverge so that they are focused on the retina.

GLAUCOMA

Glaucoma occurs when there is an abnormally high pressure inside the eyeball that is caused by a build-up of fluid. The pressure may permanently damage nerve fibres in the retina or the optic nerve, and so affect vision. In acute glaucoma, the condition develops suddenly and is accompanied by severe pain. Chronic glaucoma (see right) comes on slowly and painlessly over many years.

CHRONIC GLAUCOMA

Fluid continually moves into and out of the eye to nourish its tissues and maintain the shape of the eye. Normally, the fluid flows out through the pupil and drains out of the trabecular meshwork within the drainage angle. In chronic glaucoma, the meshwork is blocked, and pressure builds up.

